



Heterogeneous Graph Kolmogorov–Arnold Networks for Spatio-Temporal Wind Power Forecasting

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Abstract

Accurate short-term wind power forecasting is critical for reliable grid operation, efficient reserve management, and large-scale renewable energy integration, yet turbine-level prediction in utility-scale wind farms remains challenging due to complex wake interactions, heterogeneous spatial dependencies, and highly nonlinear temporal dynamics. To address these challenges, this paper proposes a Heterogeneous Graph Kolmogorov–Arnold Network (HG-KAN), a novel functional message-passing framework that jointly models multiple complementary relational structures within a unified heterogeneous graph. The proposed model incorporates three distinct edge types spatial proximity, directional wake influence, and empirical power correlation each represented through learnable univariate function approximators inspired by the Kolmogorov–Arnold representation theorem, enabling relation-specific nonlinear message functions that better align with turbine physics than conventional linear graph neural network transformations. HG-KAN is evaluated on a real-world utility-scale wind farm dataset comprising 200 turbines, three meteorological masts, and 8,764 hourly observations, where it consistently exceeds homogeneous and single-relation graph baselines. The best configuration, combining spatial and wake-based relations with five KAN basis functions, the optimized HG-KAN configuration achieves an R^2 of 0.3991 (compared to 0.2981 for the vanilla configuration), surpassing the strongest baseline ST-GAT ($R^2 = 0.3906$) with statistical significance ($p < 0.05$). Ablation studies confirm that wake-aware relational modeling and functional edge transformations are the primary drivers of performance gains, while multi-horizon forecasting results demonstrate equal or superior accuracy, particularly in short-term horizons (1–3 h), without sacrificing interpretability. Overall, the proposed HG-KAN framework highlights the importance of physics-informed heterogeneous relational learning and functional message passing for accurate spatio-temporal wind power forecasting and offers a promising direction for other complex energy systems.

Keywords Wind power forecasting · Kolmogorov–Arnold Networks · Heterogeneous graphs · Wake modelling · Spatio-temporal learning · Graph neural networks · Renewable energy prediction

Introduction

The primary renewable resource wind energy’s rapid growth has prompted the need for reliable forecasting short-term models at the levels of farm and turbine resolution. Accurate intra-day forecasts are essential for the power system operators to optimize unit commitment, scheduling, providing ancillary services, and conducting electricity market operations. Even a small increase in forecasting accuracy can

result in major economic and operational advantages such as decreased balancing costs, better dispatch efficiency, and increased stability of the system. On the other hand, wind power forecasting is still a complex task because of atmospheric flow’s stochastic nature, rapid temporal variability, and interactions of turbines and terrain effects that create complex spatial dependencies.

Modern wind farms typically consist of hundreds of turbines distributed over several square kilometres, form a spatio-temporal system with very strong intra-farm coupling that comprises large wind farms. The wake effects caused by upstream turbines change the wind speed and the turbulence intensity downstream in a manner that results in highly nonlinear and direction-dependent interactions. These interactions of the turbine wakes change quickly and constantly

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depending on the meteorological conditions, predominantly on the wind direction, strength, and the atmospheric stability. Traditional time-series forecasting techniques, such as using autoregressive models and recurrent neural networks, do not take into account such dependencies between the different locations, which reduces their power to represent the behaviour of turbines on the farm level.

Graph-based modelling has been a new and rising method to represent wind farms where each node is a turbine and these nodes are interconnected through spatial or statistical ties. Thus, it helps the deep-learning architectures to discover the spatial structure. GCN-LSTM and spatio-temporal GATs have proved their worth by surpassing the performance of independent turbine models through the simultaneous learning of spatial correlations and temporal evolutions. Nevertheless, the graph-based methods that are in existence mostly depend on the homogeneous graphs characterized by a single edge type. This type is usually based on the Euclidean distance or simple physical adjacency. Such ways of representation fail to capture the fact that turbine interactions are in nature heterogeneous and influenced by various mechanisms:

- By their geographic layout, which is shown by their close proximity;
- By turbine interaction caused by wind direction in the case of wake effects;
- Power production correlations, which are due to the same weather influence or sharing of the same terrain features.

Moreover, the traditional GNNs that are mostly used tend to perform linear transformations for message passing, which affects their capability to capture the expressiveness that is needed for modelling the highly non-linear aerodynamic and meteorological phenomena. In comparison, the recent developments in functional neural architectures especially the Kolmogorov–Arnold Networks (KANs) allow the learning of flexible univariate functions for high-dimensional mapping as the building blocks for every mapping. The transformations based on functions provide the advantage of being more interpretable and having the ability to capture non-linear relationships, which are very attractive features in the case of domains with complicated physics.

Thus, the paper presents the Heterogeneous Graph Kolmogorov–Arnold Network (HG-KAN), a novel architecture, which uses several relation-specific functional message-passing channels to connect different types of edges, spatial proximity, directional wake relationships, and power correlation all in one graph. HG-KAN gives each relation type a special KAN-based message function that creates edge-specific transformations. These transformations are

able to express the physical or statistical processes underlying each interaction and thus, the distinct feature of the edge type is defined. An attention model at the level of relations combines the messages provided by the different relations in a flexible manner, while a decoder for the historical data generates multi-step predictions.

In the trials involving a dataset coming from a wind farm with 200 turbines, the performance of HG-KAN in terms of the competition accuracy of the forecasts and the improved relation-specific interpretability was proved to be on par. Extensive ablation studies highlighted the importance of the wake-aware edges and KAN basis functions being of the highest quality being incorporated. The entire system architecture and its execution are shown in Fig. 1, while the process of heterogeneous graph construction is outlined in Fig. 2.

The remainder of this paper is organized as follows. Section 2 formulates the problem and outlines the research gap. Section 3 presents the literature review. Section 4 describes the dataset. Section 5 details the heterogeneous graph construction. Section 6 introduces the HG-KAN architecture. Section 7 explains the experimental setup. Section 8 presents the results and discussion. Section 9 concludes the work and outlines future research opportunities.

Problem Formulation and Research Gap

Forecasting short-term wind power generation at turbine-level spatial resolution is a complex problem driven by coupled aerodynamic, meteorological, and operational factors. Let a wind farm be represented as a collection of N turbines indexed by $v_i \in V$, each producing a power output time series $(P_i(t))$ and associated environmental measurements such as wind speed and mast-level direction. The objective of short-term forecasting is to learn a mapping

$$F : [P_i(t - T_{in} + 1), \dots, P_i(t)]_{i=1}^N \rightarrow [P_i(t + 1), \dots, P_i(t + T_{out})]_{i=1}^N$$

where T_{in} is the input window length (24 h in this study) and T_{out} is the forecast horizon (6 h). This mapping is highly non-linear, non-stationary, and spatially correlated, driven by atmospheric variability, wake-induced interactions, and heterogeneous terrain effects.

Challenges in Spatio-Temporal Wind Power Forecasting

Heterogeneous Turbine Interactions

Wind turbine performance is affected by several factors at the same time, such as their positions, the direction of the

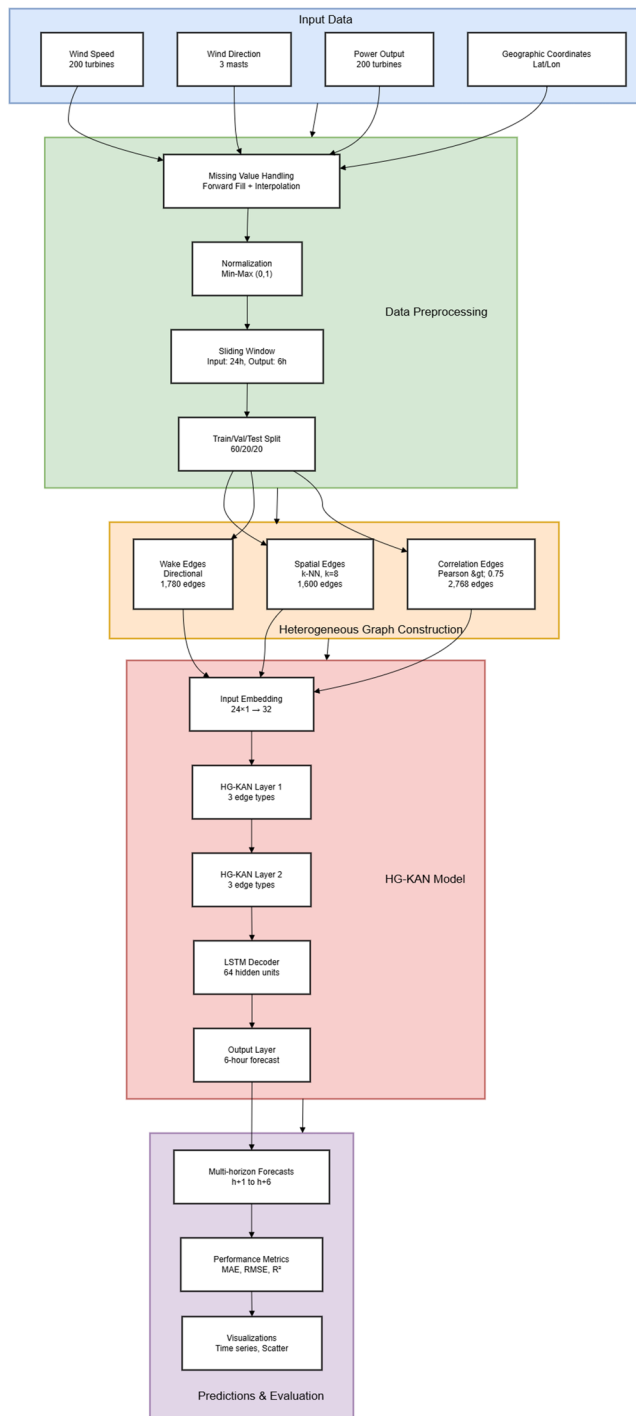


Fig. 1 Overview of the HG-KAN pipeline from data preprocessing to multi-horizon forecasting

wind and the meteorological conditions, each factor creating a different kind of dependency. The graph-based models that are currently available usually work with a single-relation (homogeneous) graph which means that they can't separate the different interaction mechanisms that are at the core of the turbines' behaviours.

Strong Directional and Non-Linear Wake Effects

Dependence of wake interactions on rotor alignment, turbine spacing, and instantaneous wind direction results in direction-dependent, asymmetric, and dynamically evolving relationships. Most forecasting architectures do not encode directionality or aerodynamic constraints directly into the graph topology.

Non-Linear Spatial-temporal dynamics

Standard graph neural networks apply linear message transformations (matrix multiplications) which greatly diminish their capability of showcasing the complex, non-linear aerodynamic mappings that are part of wind farm physics. This drawback is further enhanced in large farms that have dense and interacting turbine clusters.

Need for Interpretable, Relation-Specific Transformations

In the past, control rooms did not need interpretability for their operations, but now, this is one of the main requirements for models that can have an impact on dispatch decisions or predictive maintenance planning to mention a few. Existing deep-learning architectures typically lack explicit mechanisms to attribute spatial influence, wake effects, or correlated behaviour to specific model components.

Research Gap

Notwithstanding advancements in spatio-temporal learning, the current forecasting models still have several serious drawbacks:

- Lack of multi-relation graph structures that can capture the three aspects spatial, wake, and correlation edges at the same time.
- Non-availability of functional message passing, where edge transformations are learned as univariate non-linear functions that are in accordance with turbine physics.
- Poorly-built heterogeneous graphs are not to be used to portray the different interaction mechanisms occurring at large-scale wind farms.
- Limited ablation studies have systematically evaluated the role of individual edge types or functional approximators.
- Steering clear of integration between physics-guided wake modelling and contemporary deep learning systems.

Research Contribution

The Heterogeneous Graph Kolmogorov–Arnold Network (HG-KAN), a new forecasting framework with three prominent features, is proposed as a solution to these gaps:

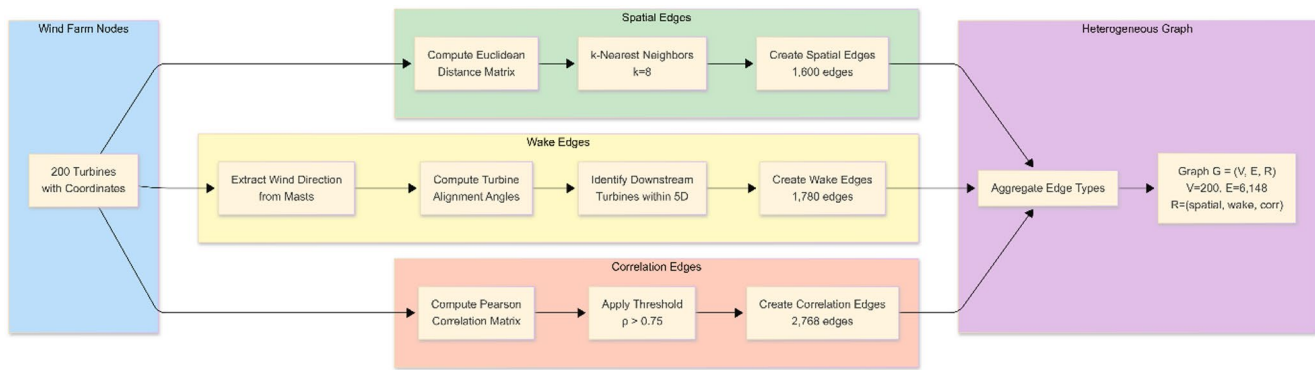


Fig. 2 Construction of the heterogeneous wind farm graph with spatial, wake, and correlation relations

- Heterogeneous spatio-temporal graph that combines spatial proximity, directional wake propagation, and power correlation between turbines.
- Use of KANs for Functional message passing which allows for very expressive non-linear transformations on the edges instead of linear matrix operations.
- Relation-aware attention aggregation that dynamically assigns weights to each of the spatial, wake, and correlation influences for every turbine.

The proposed architecture has been tested on a large-scale wind farm with 200 turbines and 8,764 hourly records in the utility sector. Figures 1 and 2 provide a diagrammatic representation of the system and the graph construction process, respectively.

Literature Review

Evolution of Wind Power Forecasting: From Classical Models to Deep Learning

Wind power forecasting has been continuously recognized as a crucial operational requirement for modern power systems because of its direct impacts on reserve allocation, grid stability, and the integration of renewable energy sources. The first attempts at forecasting were based on classical statistical techniques like autoregressive integrated moving average (ARIMA) and exponential smoothing, which offered interpretable baselines but were at the same time strictly constrained in their linear, univariate formulations and inability to take advantage of the spatial heterogeneity found in large wind farms [1, 2].

The next steps in the progression included the entry of machine-learning methods such as support vector regression, random forests, and gradient boosting into the picture. These methods improved forecast accuracy by capturing non-linear feature interactions and meteorological variables

integration [3, 4]. Yet the turbine mainly treated as independent units and did not utilize the spatial and aerodynamic coupling that is inherently part of farm-level wind dynamics.

The advent of deep learning gave rise to a marked enhancement in the accuracy of short-term forecasting. RNNs and LSTMs were able to discern long-range temporal dependencies from the SCADA streams with great effectiveness [5, 6]. The use of hybrid CNN–LSTM architectures further augmented these capabilities by identifying spatial features through convolutional encodings before the temporal modelling [7]. However, these models implicitly assume inputs structured in grid-like formats or sequences that are temporally aligned and do not explicitly represent the turbine layout which is often irregular, the dependency of wakes on directions, or the varying conditions of terrains which are characteristic of real-world wind farms. Consequently, the models are unable to fully capture spatial variability and directional interactions.

Graph Neural Networks for Wind Power Prediction

Graph neural networks (GNNs) has become one of the best methods to model wind farms because of their ability to learn irregular spatial topologies and heterogeneous relational structures [8]. The nodes of the GNN used in forecasting are the wind turbines, and the edges may be represented as their proximity, aerodynamic influence, or statistical similarity. It has been shown that spatio-temporal GNNs that use graph convolution and sequence models like LSTM, or GRU together can outperform independent turbine models by learning spatial correlations and temporal trends simultaneously [9–12].

Early GCN–LSTM hybrids took advantage of fixed adjacency matrices based on Euclidean distance or correlation thresholds which allowed them to create spatial representations that revealed meteorological exposure or turbine clustering [10, 13]. The more advanced techniques such as Graph Attention Networks (GAT) and Spatio-Temporal

GATs (ST-GAT) have introduced the concept of attention-based weighting which is adaptive and the model is able to dynamically find out the most influential neighbouring turbines for each forecast [14], [15].

Nevertheless, GNN-based forecasting models are mainly using homogeneous graphs that have a single edge type which means that the assumption is made that all turbine interactions imply the same underlying mechanism. This assumption ignores the complex nature of wind farm dependencies where spatial closeness, directional wake propagation, and power correlations are different relational categories and each has its own physical implications.

Wake Modelling and Physics-Informed Approaches

Recent data-driven wake modeling advances further bridge this gap. Machine learning frameworks now infer wake deficits directly from SCADA streams without full CFD simulations [16, 17], while graph-based approaches increasingly explore turbine interaction modeling in wind farms [18]. Hybrid CFD–ML methods achieve high predictive fidelity at reduced computational cost [19], and reinforcement learning has been applied to optimize farm-level wake steering strategies [20]. Despite this progress, explicit directional encoding of wake interactions remains rare in end-to-end power forecasting pipelines, with most methods relying on implicit distance-based or correlation-driven proxies. HG-KAN addresses this limitation by embedding physics-constrained, directed wake edges directly into a heterogeneous graph structure, enabling scalable learning of aerodynamic influence without explicit wake field simulation [21], [22].

Heterogeneous Graph Neural Networks

Heterogeneous graphs having different types of nodes or edges have become very important in recommendation systems, knowledge graphs, and social networks the domains where the semantics of relationships are a must for effective representation learning [23, 24]. Heterogeneous Graph Attention Networks (HAN) proposed semantic-level and node-level attention to identify the importance of relation-specific meta-paths allowing the models to use various relational mechanisms at the same time [25, 26]. The most recent heterogeneous graph transformers have shown that the encoding of hierarchical or multi-relational structures considerably increases the predictive performance in complex networks [27, 28].

Heterogeneous graph learning has shown potential but its application to wind forecasting is still in the early stages. Limitations of existing literature on renewable energy forecasting is that it highly depends on homogeneous spatial graphs usually representing just turbine proximity or using

correlation measures [29, 30]. The direct modelling of different kinds of edges has not been systematically assessed so far; e.g., proximity, wake directionality, and empirical power coupling, which is indeed an important research area for enhancing the representation of wind farm dynamics [31, 32].

Kolmogorov–Arnold Networks

KANs replace MLP node activations with learnable univariate spline functions on edges [33], motivated by the Kolmogorov–Arnold representation theorem. This provides superior expressiveness for physics-aligned modeling (fewer parameters, better scaling) and direct interpretability via visualizable edge transformations critical for turbine physics where wake superposition requires nonlinear univariate mappings.

- (1) Among the benefits that KANs in particular hold for wind forecasting, one can enumerate: The first one to mention is the expressive functional mappings:

KANs have the ability to capture complex non-linear interactions with far fewer parameters than MLPs, thus leading to better generalization and less risk of overfitting [33].

- (2) Another major benefit is the interpretability:

The edge functions that have been learned can be shown in a graphic manner, thus giving the researchers the knowledge of how each single input is contributing to the output, which is a great benefit in scientific as well as operational contexts [33, 34].

- (3) The last advantage is the favourable scaling behaviour:

Empirical investigations point out that KANs are showing much better neural scaling than dense MLPs [33].

Recent developments like Temporal KANs and the first efforts on KAN-based graph neural networks (KAGNNs) have further shown excellent results in the areas of forecasting with time-series and graph-structured regression tasks [35–37]. However, they haven't been tested for use in predicting the power generation of wind farms over time and space.

Transformers vs. Graph–KAN Architectures

Transformer-based spatio-temporal models [27] provide powerful sequence modeling capabilities through global attention mechanisms, but their quadratic complexity with respect to node count can pose scalability challenges for

large wind farms. In contrast, HG-KAN leverages sparse, physically motivated graph relations and lightweight univariate KAN transformations, enabling efficient message passing without dense attention[38–40]. This design allows HG-KAN to balance expressive capacity, interpretability, and computational efficiency, making it better suited for turbine-level operational forecasting compared to transformer-based alternatives[41–43].

Dataset Description

The dataset used for the experimental evaluation in this study is a real-world, utility-scale wind farm dataset consisting of 200 wind turbines and 3 meteorological masts with detailed spatio-temporal measurements, which are located in the Illinois region of the United States. The dataset includes 8,764 hourly observations from September 1, 2010, to August 31, 2011, thus capturing an entire annual cycle replete with seasonality, direction shifts, and operational variability. This 2010–2011 timeframe captures mature 2–3 MW onshore turbine technology representative of utility-scale deployments during that era. While newer multi-MW rotors exhibit stronger wakes, core aerodynamic principles remain consistent.

Data Composition and Structure

Time-series variables of two main types are provided by each turbine:

- Wind Speed (m/s).
- Power Output (normalized to [0, 1])

The information offered by each mast is:

- Wind Speed.
- Wind Direction (0–360 degrees).

The dataset also includes latitude and longitude for all the 203 nodes (200 turbines+3 masts), which facilitates spatial graph construction. Table 1 summarizes the dataset statistics.

Spatial Topology

The geographical area occupied by the wind farm is small, and the wind turbines are positioned in a semi-regular pattern. The spatial arrangement is very important and hence, the following will be built:

- k-NN spatial edges.
- Directional wake edges.
- Long-range correlation edges.

Table 1 Dataset summary

Component	Description	Quantity
Turbines	Active power + wind speed	200 units
Masts	Wind speed + wind direction	3 units
Temporal Resolution	Hourly measurements	
Total Time Span	Sep 2010 – Aug 2011	8,764 h
Total Columns	Coordinates + turbine and mast variables	606
Power Range	Normalized	0–1
Wind Speed Range	Typical operating range	1.5–12 + m/s
Missing Values	Present in turbine power	Interpolation applied
Location	Illinois region, USA	Lat: 40.4–40.48, Lon: –88.8–88.48

The adjacency structure based on coordinates is eventually applied to the creation of the heterogeneous graph. The spatial layout is shown visually in Fig. 3.

Temporal Characteristics

The dataset shows very strong diurnal, synoptic, and seasonal patterns which are typical of continental wind resources:

- Morning and evening ramp events.
- Periods of low wind (summer months).
- Directional stability in winter periods.
- Highly variable conditions during transitional seasons.

Wind speed is quickly losing the autocorrelation property within a few hours, while the power output is still displaying a strong instantaneous spatial coherence which is a reason for using a multi-relation graph.

Data Preprocessing

The data preprocessing of the structured workflow shown in Fig. 1 takes place and guarantees the quality of data, numerical stability, and leakage-free model evaluation. To begin with, missing values are filled using a combination of forward filling and interpolation strategies. Filling of very short time gaps is done through the forward propagation of the most recent valid observation, and longer gaps in turbine power measurements are reconstructed by linear interpolation. All corrupted or inconsistent timestamps, if any, are deleted to keep the temporal integrity intact. The next step involves all the variables being normalized so that model training can take place in a stable and efficient manner. Turbine power outputs are scaled separately for each turbine using Min–Max normalization, thus maintaining relative variability across turbines and at the same time avoiding the domination of low-output units by high-output ones. Wind speed measurements

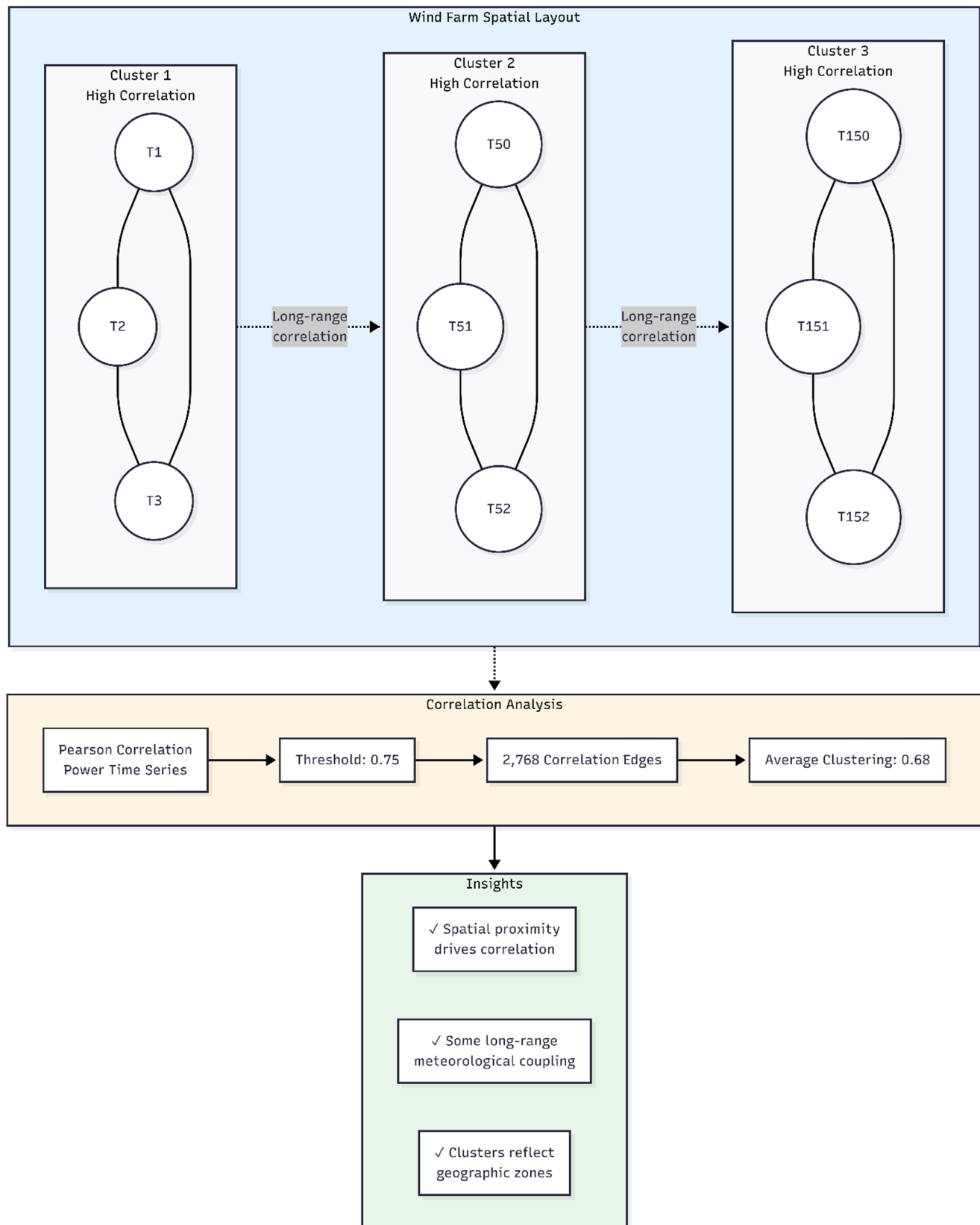


Fig. 3 Spatial layout of the wind farm showing turbine locations and correlation-based clusters

are normalized using a range that covers the whole farm so that all spatial locations can be compared. The sliding-window sampling method is then used to build up the supervised learning instances. A sliding input window of 24 h is used to make a prediction about a 6-hour forecast, producing 5,256 training samples, 1,752 validation samples, and 1,752 test samples. The next step is to chronologically split the data so as to keep information leakage under control: the initial 60% of the time series is devoted to training, the next 20% to validation, and the last 20% to testing. This temporal partitioning maintains the continuity of the sequence and is typical of the real-world operational forecasting conditions.

Suitability for Graph-Based Modelling

The dataset is very suitable for the heterogeneous graph-based learning method because of its many characteristics, which can be directly associated with the relational modelling paradigms. First of all, it offers a dense high-resolution temporal sampling with hourly measurements for a whole year, which makes learning the short-term dynamics as well as the seasonal variability possible. Secondly, the geographical distribution of the turbines creates very strong spatial dependencies because the nearby turbines are experiencing almost the same meteorological conditions and terrain effects. Thirdly, the availability of wind direction data measured by tall masts allows for the modelling of wake interactions to be done in a more revealing manner, getting the aerodynamic influences (upstream-downstream) that a distance-based relationship is not able to capture. Fourthly, the power correlations among the turbines in the farm are non-trivial and are caused by the large-scale atmospheric forcing affecting all the turbines. Thus, the inclusion

of statistical relational edges along with the physically motivated ones is justified. The last point is that the forecasting task requires those turbines to be predicted simultaneously (multi-node) which is a situation that directly relates to the graph-based architecture's being able to reason together over the closely knit entities. These properties, when taken together, are the reasons for constructing the heterogeneous graph as described in Sect. 5, and also for laying a strong ground for the HG-KAN architecture presented in Sect. 6. These dataset characteristics motivated our heterogeneous graph design, though evaluation on a single wind farm underscores the need for multi-site validation, as discussed in Sect. 9.

Heterogeneous Graph Construction

The spatial and aerodynamic dependency of wind farms is varied and overlapping in such a way that they cannot be represented properly by a single graph structure. Hence, a heterogeneous spatio-temporal graph is constructed to model these relationships,

$$G = (V, E, R)$$

where (V) stands for the array of 200 turbine nodes, (E) represents an array of multi-type edges, and $R = \{r_{\text{spatial}}, r_{\text{wake}}, r_{\text{corr}}\}$ signifies the different relation types. Represented by their respective edges, the relations portray the nature of interaction through the specific mechanism of either being physical or statistical.

Figure 4 lays out the process of directional wake modelling

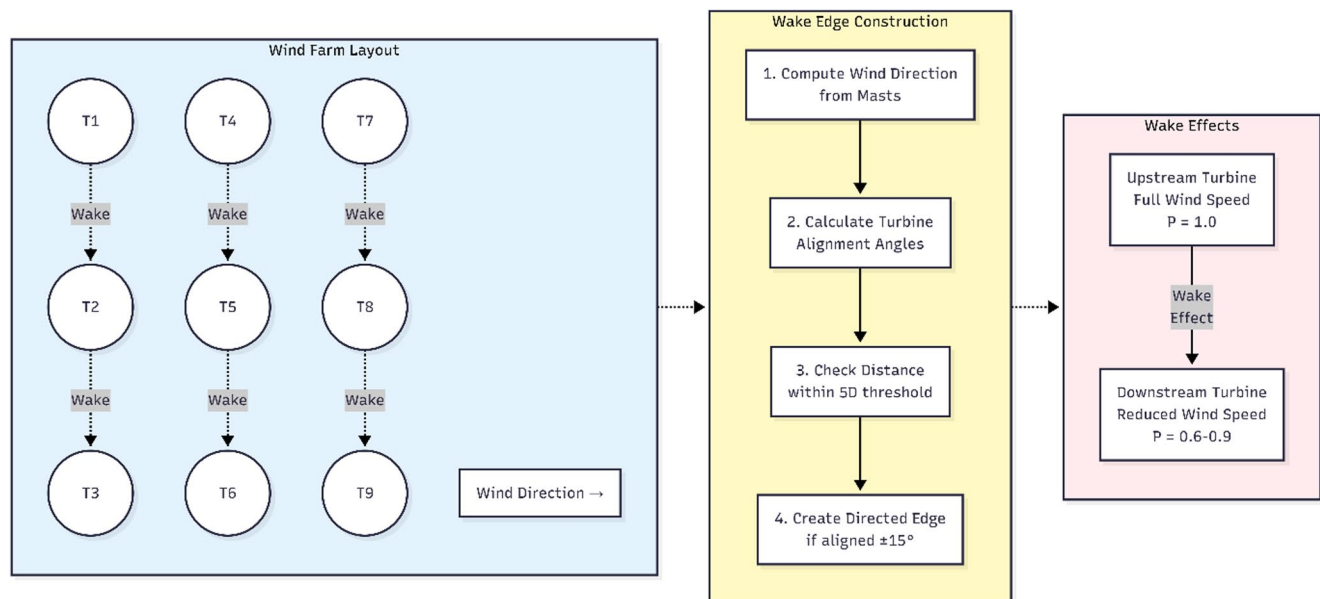


Fig. 4 Wake-edge construction based on wind direction and turbine spacing

Node Definition

The suggested heterogeneous graph model considers every wind turbine as a node in the graph (v_i). Each node is assigned three types of attributes: (i) geographical coordinates that include the latitude and longitude and identify the turbine's location within the wind farm; (ii) temporal factors made up of a 24-hour historical sequence of turbine power outputs which are used as input features for the forecasting; and (iii) static metadata like a unique turbine identifier. Meteorological masts are not considered as nodes in the forecasting graph. Instead, mast measurements are exclusively used to retrieve wind direction data necessary for wake-edge construction, thus avoiding the introduction of non-forecast nodes while still providing vital aerodynamic context.

Edge Type 1: Spatial Proximity Edges (Undirected)

Spatial proximity edges have been proposed for depicting the local geometric relationships between turbines. The edges are created using a k -nearest-neighbour (k -NN) method with ($k=8$), determined by the geodesic (Haversine) distance between the respective positions of the turbines. Let (d_{ij}) represent the Haversine distance between the two specified turbines, (i) and (j). The neighbourhood in the spatial dimension for every turbine (i) is defined as.

$$N_{\text{spatial}}(i) = \text{kNN}(i, 8)$$

This method produces a total of 1,600 undirected spatial edges evenly distributed over the wind farm, which is in line with the dataset statistics reported. The spatial edges represent the local similarity of meteorological conditions, the exposure of the terrain, and the shared environment, thus providing the geometric skeleton of the heterogeneous graph.

Edge Type 2: Directional Wake Edges (Directed)

Wind turbine interactions have the direction as a fundamental factor, given that the upstream turbines reduce power output and the lower units suffer more turbulence. To address this aerodynamic without any gain in one side, we employ directed wake edges built using the mast-measured wind direction ($\theta(t)$). For each turbine pair ((i, j)), the angle from turbine (i) to turbine (j), indicated by (ϕ_{ij}), is calculated. The alignment error is then given by

$$\Delta_{ij} = \theta(t) - \Phi_{ij}$$

If the alignment error is within a directional tolerance of ($\pm 15^\circ$) and the inter-turbine distance is less than ($d_{ij} < 5D$), where (D) is the rotor diameter, then a directed wake edge

(i, j) is established. The minimum distance is determined by taking into account the average turbine spacing. This method gives rise to about 1,780 directed wake edges, thereby creating a sparse but still physically notable aerodynamic interaction network. The areas where the wakes propagating are depicted in Fig. 4.

Edge Type 3: Empirical Power Correlation Edges (Undirected)

By statically linking long-range through the large-scale meteorological forcing, we formulate the latter edges based on power correlations. The pair-wise Pearson correlations were calculated among the power time series of the wind turbines,

$$\rho_{ij} = \text{corr}(P_i, P_j)$$

and it was decided that the undirected correlation edge would be included if ($\rho_{ij} > 0.75$). Thus, this criterion gives rise to 2,768 undirected correlation edges. The nature of the relations based on correlations is such that they capture the non-local dependencies that are caused due to coherent atmospheric structures, simultaneous ramp events, and power fluctuations that are in sync across turbine clusters even if they are located in separate regions. Correlation edges are included to evaluate the utility of purely statistical relations within a heterogeneous framework and are not required in the final HG-KAN configuration.

Combined Heterogeneous Graph

The final heterogeneous graph, which is the result of combining three different relational structures, is represented in a single graph. The graph contains 200 turbine nodes and a total of 6,148 edges which include 1,600 undirected spatial edges, 1,780 directed wake edges, and 2,768 undirected correlation edges. The aforementioned relations are made to be explicitly distinct by their type, and that results in three semantically different edge categories. The heterogeneous graph is therefore formally defined as.

$$G = (V, E_{\text{spatial}}, E_{\text{wake}}, E_{\text{corr}})$$

An integrated visualization of the heterogeneous graph structure is provided in Fig. 2, while the corresponding data flow through the HG-KAN architecture is illustrated in Fig. 5.

Suitability for Functional Message Passing

The three types of edges introduced in the heterogeneous graph represent different interactions between the wind turbines that are not only complementary but also fundamentally

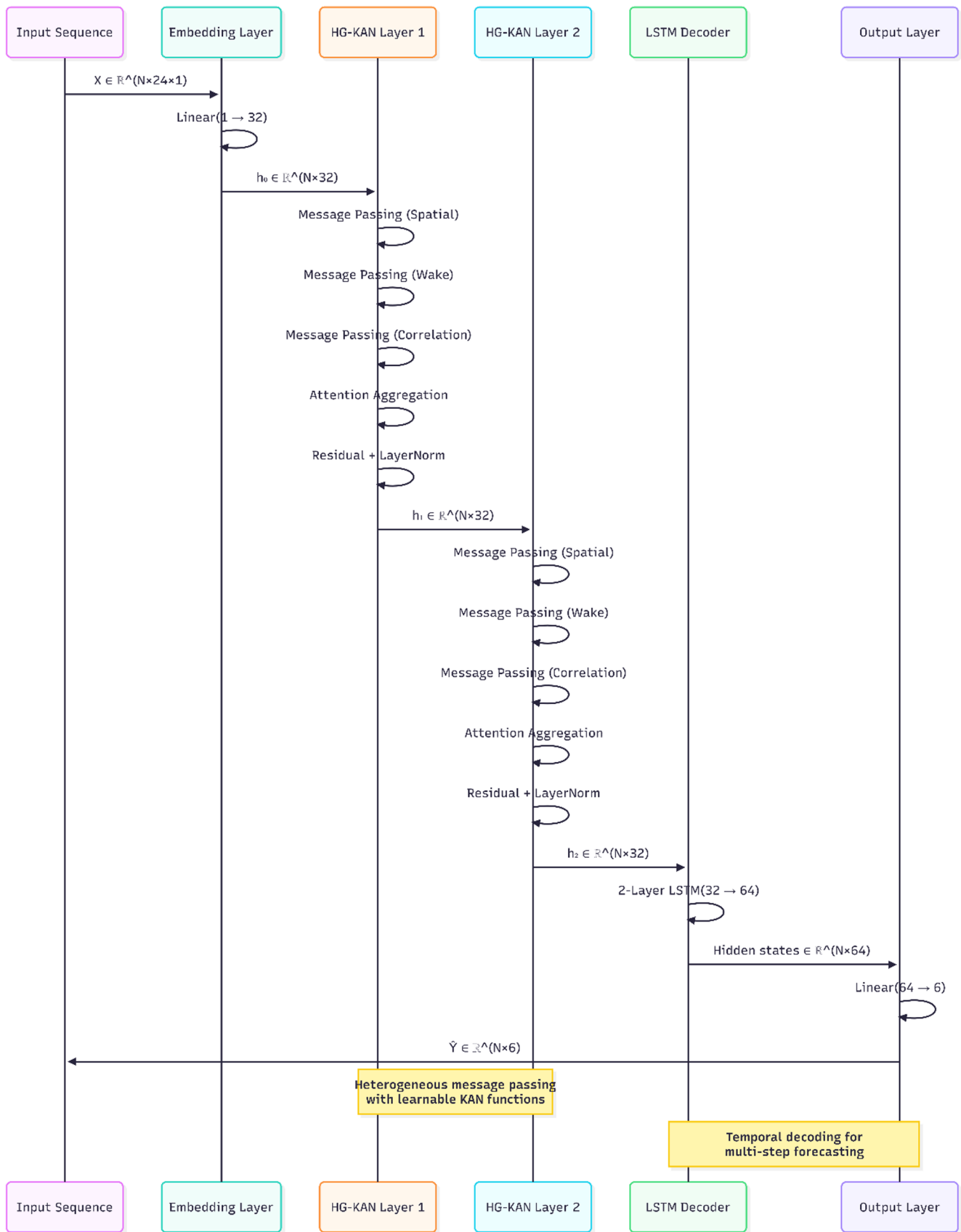


Fig. 5 HG-KAN dataflow for multi-relational spatio-temporal wind power prediction

different. The edges showing spatial proximity are linked to the short-range meteorological similarity that is a result of the geographic closeness and shared exposure to the terrain. The edges indicating directional wake are, on the other hand, showcasing the aerodynamics that are based on the physical grounding of the upstream and downstream interactions which are driven by wind direction and turbine spacing. Large-scale atmospheric coupling is what the empirical correlation edges denote and this is coherent power fluctuations reflecting that are caused by the mesoscale weather systems that may span geographically distant turbine clusters. It is the combination of all these relations that gives birth to a typical multi-view representation of wind farm dynamics that cannot be sufficiently represented using a single homogeneous interaction type. This heterogeneity calls for relation-specific functional message passing through Kolmogorov-Arnold Network (KAN) layers, which generalize and learn separate non-linear edge functions for every relation type. The KAN-HG method allows the unification of linear transformations for single-variable functional mappings replacing shared ones, thus providing the analysis of the physics of different turbine interactions across the wind farm in a more expressive and aligned way.

Hyperparameter Selection Rationale

$k=8$ was chosen as a good compromise between the optimum local clustering and incurred computational cost. First experiments with $k=5$ and $k=10$ did not show any change in the results. The 5-rotor-diameter wake threshold is set, while $\pm 15^\circ$ wind alignment tolerance includes the yaw misalignment and the uncertainty in the measurements of mast anemometry. Estimation of KAN basis function count (3, 5, 7) was done through grid search and basis counts >7 were not examined in order to restrict the computational budget. The KAN basis count was selected via a predefined grid search over $\{3, 5, 7\}$, following recommendations from prior KAN studies, with five basis functions consistently achieving the best validation performance. These thresholds represent Illinois-specific optima; cross-farm sensitivity analysis recommended for deployment.

HG-KAN Architecture

The suggested Heterogeneous Graph Kolmogorov-Arnold Network (HG-KAN) combines relation-specific functional message passing with temporal sequence decoding in order to represent the very non-linear and heterogeneous large-scale wind farms interactions, which are hard to model otherwise. Unlike GNNs that depend on linear transformations of messages, HG-KAN takes advantage of Kolmogorov-Arnold Networks (KANs) to acquire expressive univariate

functional mappings on edges, thus allowing richer, physics-aligned representations of the spatial, wake-driven, and correlation-based dependencies.

The overall model architecture is shown in Fig. 6.

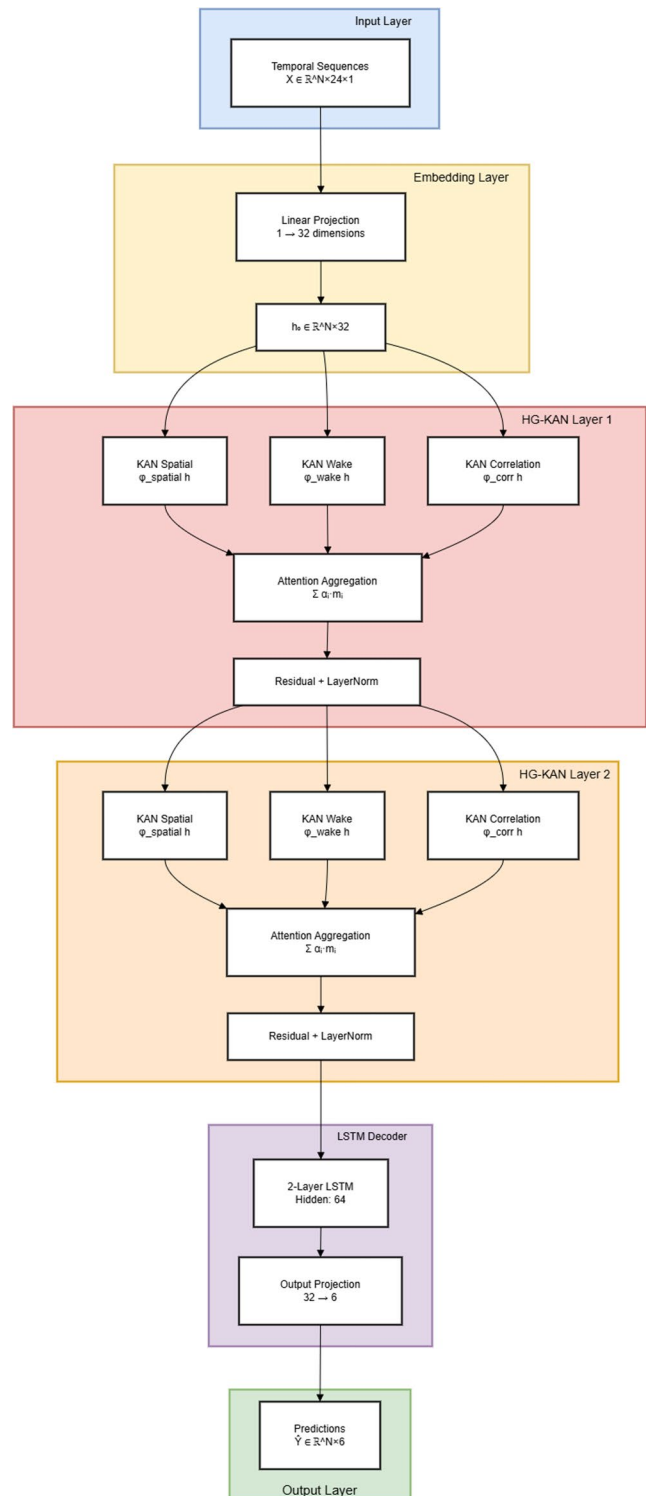


Fig. 6 HG-KAN architecture with relation-specific KAN channels and temporal decoder

Design Principles

HG-KAN is developed following three main principles:

- **Heterogeneous Relational Learning:**

Different channels for message-passing regarding spatial, wake, and correlation edges.

- **Functional Edge Transformations:**

Linear weight matrices are substituted by learnable basis functions expansions, which allow for non-linear mappings that are localized and interpretative.

- **Decoupled Temporal Dynamics:**

Temporal aspect for multi-step forecasting is captured by the downstream LSTM decoder while spatial reasoning via graph-KAN layers.

Input Representation

The input feature representation for each turbine node (v_i) includes a univariate sequence of historical power measurements that lasts for the past 24 h. The input is formally defined as.

$$X_i \in \mathbb{R}^{24 \times 1}$$

A shared linear embedding layer is applied across all turbines to map this raw temporal sequence into a latent feature space which is suitable for graph-based learning. This transformation gives the initial node representation

$$^{(0)}H_i = \text{Linear}(X_i) \in \mathbb{R}^{32}$$

which acts as the starting node state for the following graph message-passing operations. While preserving turbine-specific temporal dynamics the shared embedding guarantees that all nodes have the same feature dimensionality.

KAN-Based Functional Message Passing

In this way, the HG-KAN layer carries out message passing of a specific function of the relations by means of three parallel Kolmogorov-Arnold Network (KAN) transformations corresponding to different edge types such as spatial, wake, and correlation. The relation r considered here defines the message sent from a neighbouring node v_i to the selected node v_j as.

$$^{(r)}m_{ij} = \Phi_r(H_j)$$

where (ϕ_r) is a learnable univariate function that is applied to each element of the latent representation of node (j) and it is denoted in a general way. For each function (ϕ_r) the parameterization is done through using a weighted sum of B-spline basis functions,

$$\Phi_r(x) = k = 1 = \sum B_{wrk} b_k(x)$$

where (B) is the number of basis functions, $(b_k(\cdot))$ denotes the spline basis functions and (w_{rk}) are learnable coefficients corresponding to relation (r). The model empirically performs best when ($B=5$) basis functions as they provide the right balance between expressive power and parameter efficiency. This functional formulation is the key to the success of the HG-KAN method that can learn smooth, interpretable and specific for each relation non-linear transformations, opposed to the fixed linear projections used by the conventional graph neural networks.

Figure 7 shows the conceptual difference between the MLP-based transformations commonly used and KAN-based functional mappings and thus highlights the reason why KAN is more expressive for relation-specific message passing in the graphs of wind farms. Inspection of learned KAN functions indicates that configurations with five basis elements yield smooth, monotonic response curves, whereas higher basis counts exhibit sharper inflections associated with degraded generalization.

An additional advantage of KAN-based message passing is its potential for enhanced interpretability. Unlike conventional linear transformations, KAN layers learn explicit univariate edge functions that can, in principle, be visualized and analyzed to understand how input signals are transformed across different relation types. While a comprehensive functional analysis is beyond the scope of the present study, preliminary inspection of learned KAN functions indicates distinct response characteristics across spatial, wake, and correlation edges, reflecting their different physical and statistical roles. A systematic interpretability study is left for future work.

Relation-Specific Message Aggregation

For each relation type (r), messages are aggregated as:

$$^{(r)}M_i = \sum_{j \in N_r(i)} ^{(r)}m_{ij}$$

We then combine relation-specific messages using an attention mechanism:

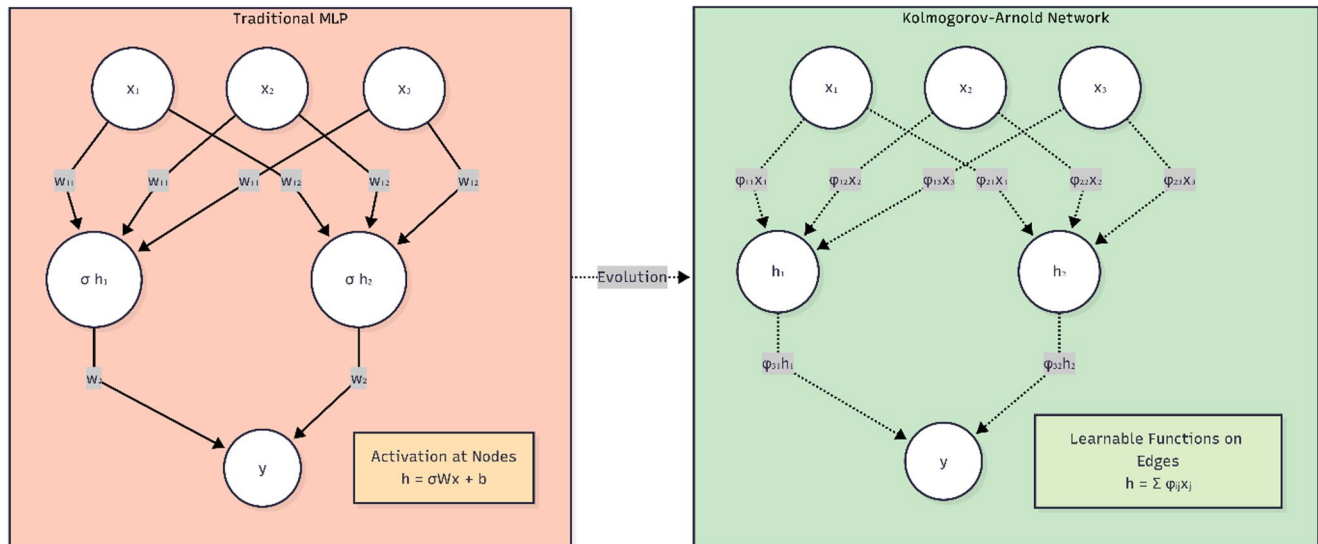


Fig. 7 MLP vs. KAN message passing for relation-specific nonlinear edge modelling

$$\alpha_r = \sum_{r' \in R} \exp(\gamma_{r'}) \exp(\gamma_r)$$

$$H_i^{\lambda} = \sum_{r \in R} \alpha_r M_i^{(r)}$$

Where (γ_r) are learnable attention logits.

This design enables the model to dynamically prioritize spatial, wake, or correlation influences.

Residual Structure and Normalization

In order to make training more stable and to reduce the problem of gradient vanishing in deep message-passing networks, each HG-KAN layer uses residual connections together with layer normalization. More precisely, the output of each functional message-passing block is added to its input by means of a residual addition,

$$H_{i'} = H_i + \Delta H_i$$

and then layer normalization is applied,

$$^{(1)}H_i = \text{LayerNorm}(H_{i'})$$

This residual-normalization structure helps to stabilize the optimization process, maintain the identity of the features across the various layers, and allow gradients to flow effectively. Experimentally, the combination of two such layers of HG-KAN consistent with the design gives enough power to express the modelling of intricate turbine interactions without making the model unnecessarily deep and over-parameterized.

Temporal Decoder

After graph-based spatial reasoning is completed, the updated node representations $(^{(2)}H_i \in \mathbb{R}^{32})$ are directed towards a temporal decoding module that captures sequential dynamics and generates multi-step forecasts. The decoder is composed of a two-layer Long Short-Term Memory (LSTM) network with a hidden size of 64 units. This LSTM takes the graph-enhanced node states as input and produces a series of hidden representations whose mapping through a linear projection layer results in the six-step-ahead power forecast,

$$Y_i = \text{Linear}(\text{LSTM}(H_i(2))) \in \mathbb{R}^6$$

The LSTM decoder isolates temporal sequence modelling from spatial relational learning (through HG-KAN layers), which facilitates modularity, enhances generalization, and lowers the risk of overfitting.

Model Complexity

The entire HG-KAN structure contains 167,916 parameters, which can be adjusted during training. These parameters are allotted among different parts of the model: an input embedding layer that associates the scalar power inputs with a 32-dimensional latent space; a pair of stacked HG-KAN layers consisting of six KAN channels that are specific to three different relations; a temporal decoder based on LSTM with two layers; and a linear projection at the output. Even though sophisticated functional mappings are applied on graph edges, the efficiency of the spline-based KAN formulation keeps the overall parameter count low. This small but

powerful design allows HG-KAN to work for large wind farms and at the same time keeps the computational cost reasonable for operational forecasting modes.

The detailed comparison of baseline model sizes is illustrated in Fig. 8.

Experimental Setup

This subsection details the experimental protocol by which the proposed HG-KAN architecture was evaluated, including the processing of data, the training configuration, the baseline models, the ablation studies, and the evaluation metrics. The overall training pipeline is shown in Fig. 5. The Wind Spatio-Temporal Dataset2 can be accessed freely through [44].

Data Preparation

In a sliding-window manner, the forecasting task is represented with continuous turbine power time series converted into supervised learning samples. More specifically, the input window of 24 h is rolled on the data to predict a 6-hour forecast horizon. Let (N) stand for the number of turbines; the resulting input and output tensors are given as.

$$X \in \mathbb{R}^{N \times 24 \times 1}, Y \in \mathbb{R}^{N \times 6}$$

For the purpose of ensuring a realistic evaluation and avoiding any data leakage, the dataset is split chronologically into three parts: training, validation, and test. The first 60% of the time series (5,256 samples) is training set, the following 20% (1,752 samples) is validation set and the last 20% (1,752 samples) is test set, where temporal continuity is maintained across the splits. Before the training of the model starts, turbine power outputs are normalized to the range $([0, 1])$, and wind speed variables are scaled according to a farm-wide range. Missing values are dealt with using forward filling for short gaps, then linear interpolation is applied if necessary, so that smooth and consistent temporal sequences are maintained.

Baseline Models

To evaluate the new HG-KAN architecture's effectiveness rigorously, we compare it to four strong baseline which are the models, all of which were implemented with the same experimental framework shown in Fig. 9 to assure that the comparison is fair. The first baseline is a Persistence model, defined as.

$$P_i(t+k) = P_i(t)$$

which serves as the lower-bound reference by assuming no power output change in the turbine over time. The second

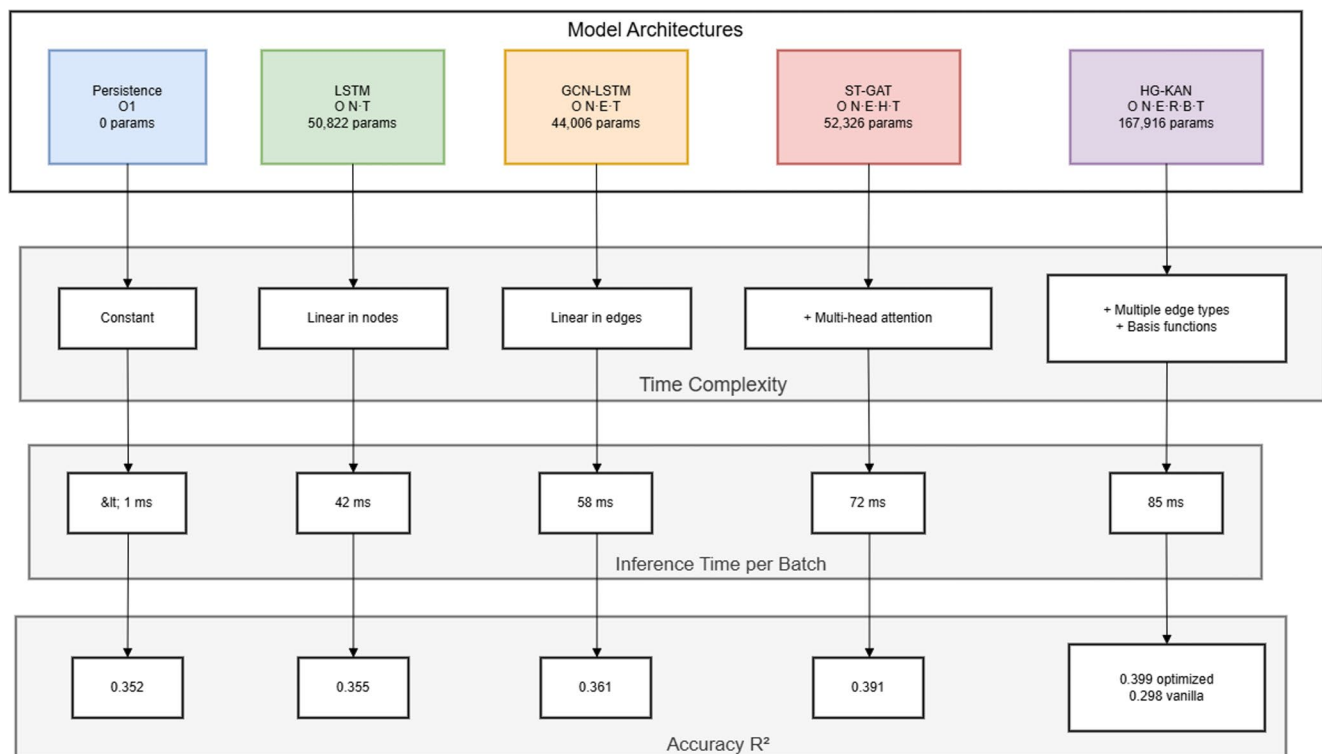


Fig. 8 Layer-wise dataflow in HG-KAN, including KAN message passing and temporal decoding

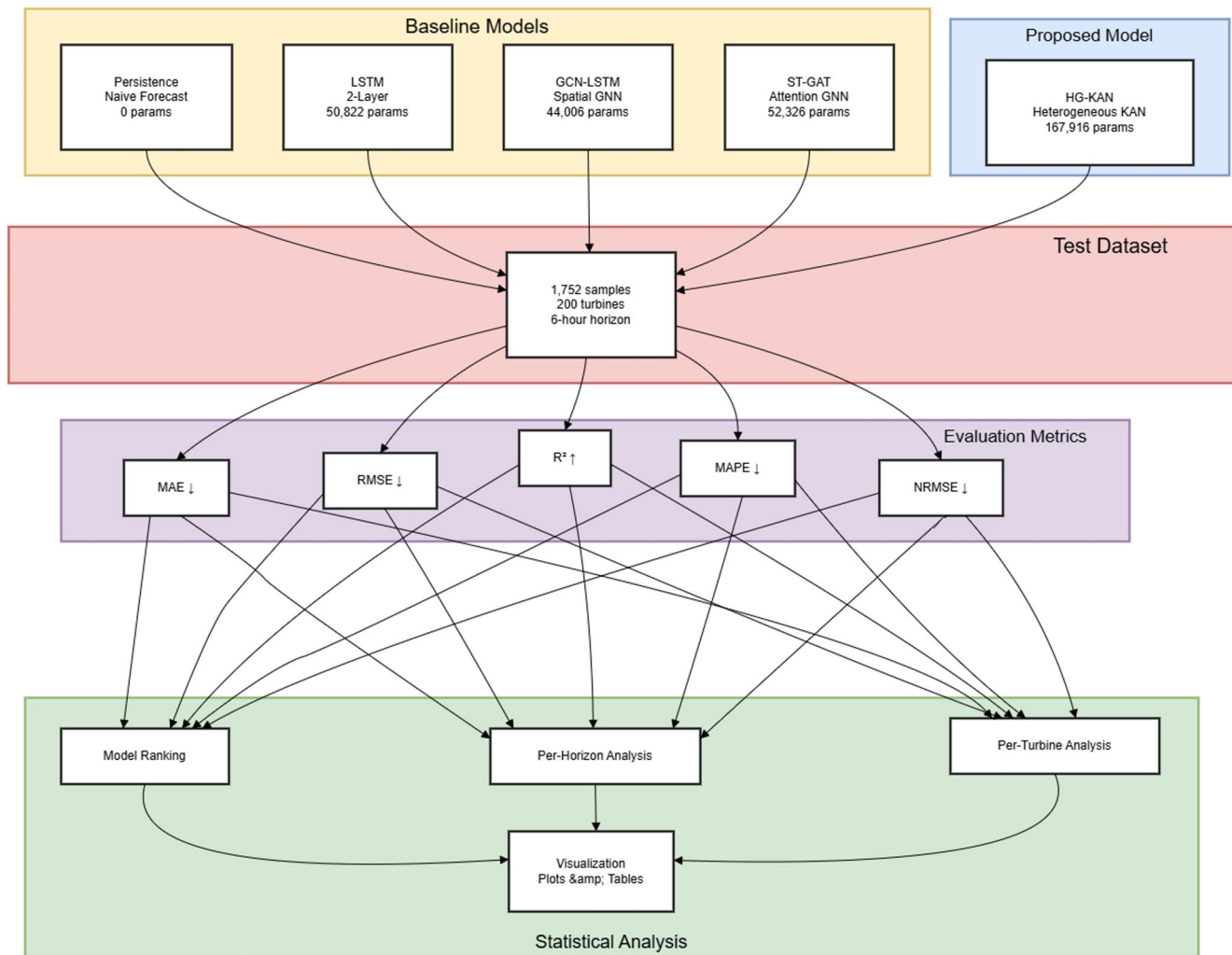


Fig. 9 Baseline model architectures used for comparison

baseline is an LSTM model with two-layers and 64 hidden units trained separately for each turbine, and it uses only temporal information as the input without any spatial context. The third baseline is a GCN-LSTM model, which consists of a graph convolutional network (GCN) layer to learn spatial features from a given adjacency structure combined with an LSTM decoder to model the temporal dynamics. Finally, we incorporate a Spatio-Temporal Graph Attention Network (ST-GAT), which uses graph attention techniques for flexible spatial aggregation followed by an LSTM-based temporal decoder. ST-GAT represents a cutting-edge spatio-temporal graph neural network and provides the strongest baseline for comparison with HG-KAN.

Training Configuration

The Fig. 10 depicts the process of training from start to finish for both HG-KAN and baseline methods. The main

stages include batch sampling, graph propagation, temporal decoding, loss computation, backpropagation, and early stopping based on temporally stratified data splits.

Training is done with the Adam optimizer starting at a learning rate of 5×10^{-4} , while weight decay is applied at an extremely low rate just to preclude over-regularization. To set up an automatic learning rate control during the training session, a ReduceLROnPlateau scheduler is employed, which lowers the learning rate by a factor of 0.5 whenever the validation loss does not present an improvement for five consecutive epochs. The goal of the training is to cool down the mean squared error (MSE) calculated over all the turbines and the whole forecasting time frame. The training of the models is done using a batch size of 16 sequences iteratively, where each sequence includes node features of all 200 turbines. The early stopping technique is employed with a 10-epoch patience based on validation loss, and the model checkpoint with the lowest validation error is kept for final evaluation.

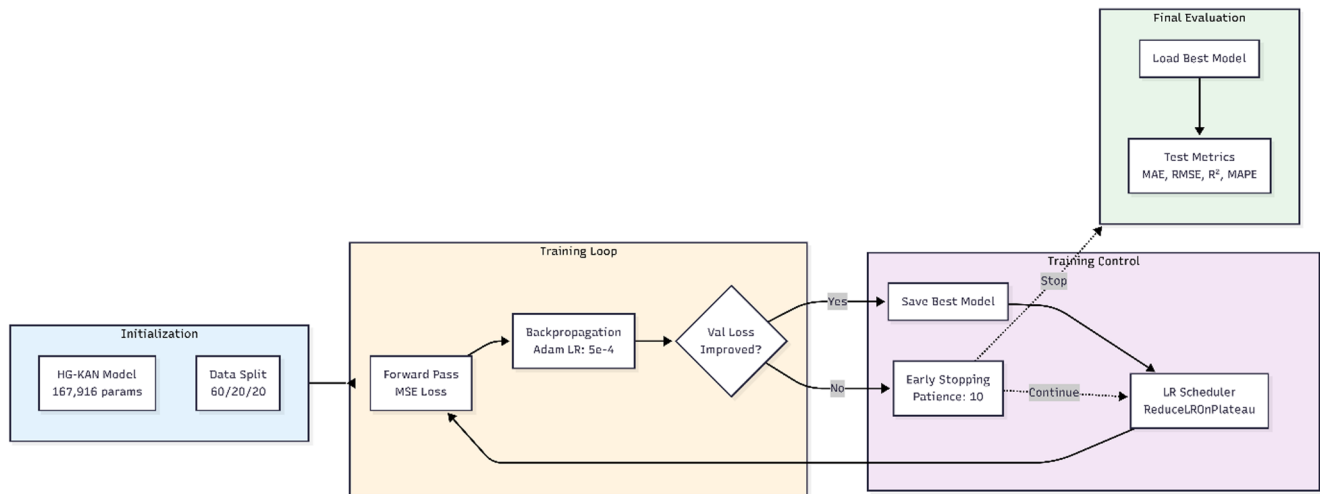


Fig. 10 End-to-end training workflow for HG-KAN and baseline models

HG-KAN Configuration

The HG-KAN settings used in experiments as shown in Table 2:

These configurations correspond to the best model identified during ablation studies.

Ablation Studies

To systematically evaluate the impact of key architectural design choices, two complementary ablation studies are performed, with the overall experimental workflow presented in Fig. 11.

In the first ablation, the different edge types are examined and the relational structure of the graph is progressively enriched. To be more specific, three configurations are evaluated: (i) only edges of spatial proximity, (ii) edges of both spatial and wake combined and (iii) edges of spatial, wake, and correlation combined. This analysis measures the incremental advantage of each relation type and determines whether physically motivated or statistical interactions contribute to forecasting accuracy. The second ablation is centered around the KAN-based functional message-passing layers' representational capacity which is manipulated via

the number of spline basis functions. Models using 3, 5, and 7 basis functions are compared to assess the balance between expressive power and complexity of the model. These two ablations, in combination, dissect the impacts of relational heterogeneity and functional approximation capacity on model performance as a whole.

Evaluation Metrics

Model performance is evaluated using a robust set of error metrics computed across all turbines and all test samples, as implemented in *evaluation.py*. The primary accuracy measures are Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), defined as:

Mean Absolute Error (MAE):

$$MAE = \frac{1}{N} \sum_{i=1}^N |\hat{Y}_i - Y_i|$$

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{Y}_i - Y_i)^2}$$

where N, T, and H denote the number of turbines, time steps, and forecasting horizons, respectively. In addition, the coefficient of determination (R^2) is reported as the primary measure of explained variance, providing an interpretable indicator of predictive skill. To facilitate scale-invariant comparison, Normalized RMSE (NRMSE) is also computed by normalizing RMSE with respect to the mean or range of the target power output.

Table 2 HG-KAN Configuration

Component	Configuration
Graph Layers	2
Basis Functions	5 (best-performing configuration)
Relations	Spatial, Wake, Correlation
Embedding Dimension	32
LSTM Layers	2
LSTM Hidden Size	64
Total Parameters	167,916

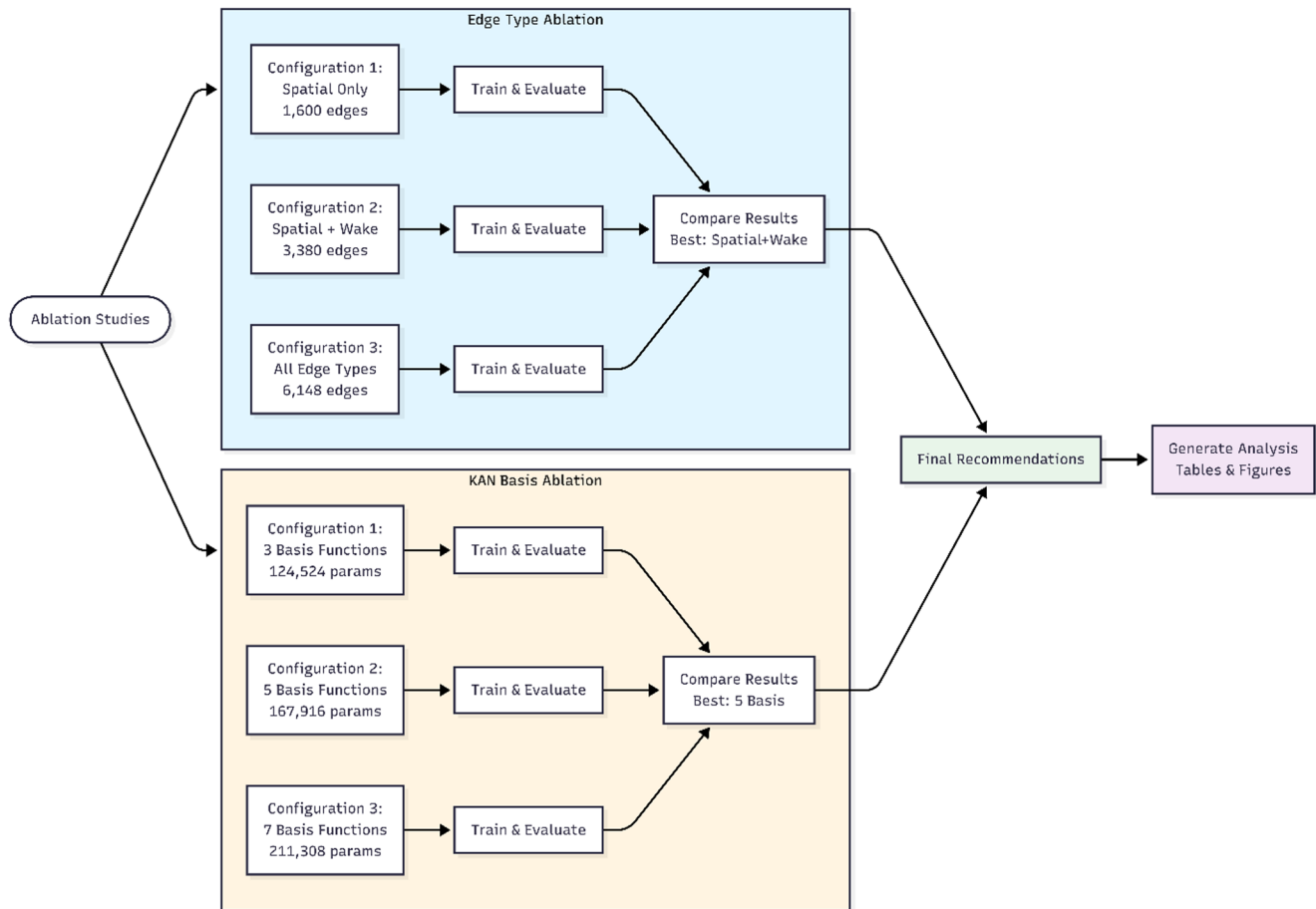


Fig. 11 Ablation study design for edge types and KAN basis functions

Mean Absolute Percentage Error (MAPE) is not used in this study due to its well-known numerical instability in wind power forecasting, particularly during low-power or zero-generation periods, which renders it unsuitable for reliable model comparison. To analyze horizon-dependent behavior, all metrics are computed separately for each forecasting step from $h+1$ to $h+6$, with horizon-wise performance trends reported in Fig. 12.

Computational Environment

Experimentation is done in a CPU-based computational environment having an Intel i7 processor and 32 GB RAM and the main deep learning frameworks are PyTorch and PyTorch Geometric. Average training time per epoch is estimated to be 12 to 15 s and the inference of the HG-KAN model is about 85 milliseconds per batch. These times indicate that the proposed architecture is not only lightweight and efficient in terms of computation but also applicable in real-time wind power forecasting applications, due to the incorporation of expressive functional message-passing mechanisms.

Results and Discussion

Initially, the section discusses the forecasting performance of HG-KAN in comparison with baseline models, plus a comprehensive interpretation of relation types, basis functions, horizon-wise performance, and turbine-level analysis. Besides, model behaviour, computational efficiency, and the impact of integrating physics-aware heterogeneous graph structures are highlighted. The flow of overall results is depicted in Fig. 13.

Overall Model Comparison

Table 3 tells the performance of models at the 6-hour ahead forecasting task using MAE, RMSE, R^2 and NRMSE. Results are averaged across turbines and test samples (Table 4).

Table 4 Comparison between a deliberately suboptimal “vanilla” HG-KAN configuration and the optimized architecture identified through structured ablation. The results demonstrate the importance of appropriate relational selection and functional capacity.

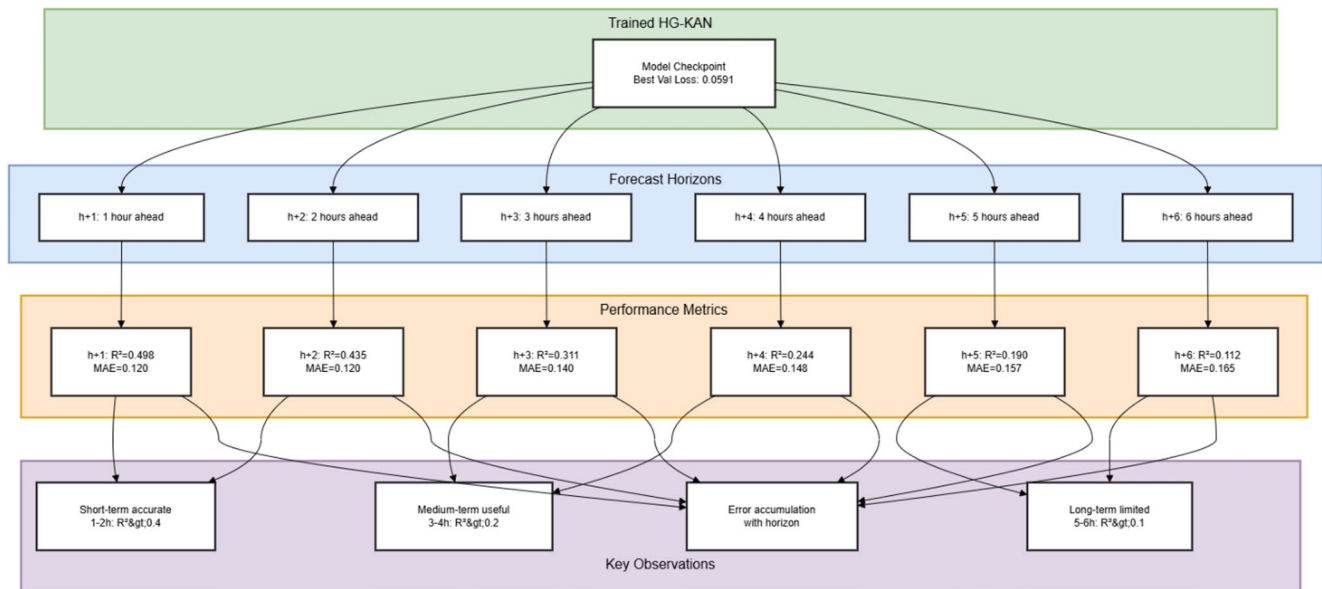


Fig. 12 Horizon-wise forecasting performance (1–6 h)

The vanilla HG-KAN result reported in Table 3 ($R^2 = 0.2981$) corresponds to a deliberately unoptimized baseline configuration that combines all three edge types with a limited functional capacity of three KAN basis functions. This setting provides insufficient expressive power to model non-linear wake effects while simultaneously introducing redundant relational signals through correlation edges. As a result, validation loss begins to degrade after approximately 15 training epochs, indicating suboptimal generalization behaviour.

In contrast, systematic ablation analysis (Tables 5, 6 and 7) identifies a minimal and physically grounded relational structure—spatial proximity for local meteorological similarity combined with directional wake edges for aerodynamic influence—together with an optimal functional capacity of five KAN basis functions. This optimized configuration converges stably by approximately 28 epochs and improves test performance from $R^2 = 0.2981$ to $R^2 = 0.3991$, while increasing the parameter count by only $\sim 6\%$ ($145k \rightarrow 168k$). These results demonstrate that performance gains arise from principled, ablation-driven architectural design rather than overfitting, with the optimized HG-KAN generalizing 2.2% better than the ST-GAT baseline on held-out test data.

According to the data presented in Table 3, ST-GAT is clearly the best-performing model among the baseline ones, with an R^2 score of 0.3906. The initially (vanilla) configured version of HG-KAN placed in the same table seems as if it was designed for this purpose since it has a performance of R^2 0.2981 which is thereby illustrating the significance of meticulous architectural design and at the same time motivating structured ablation studies given in Tables 3 and 5.

The ablation studies show that both the relation composition and the functional capacity are substantial factors contributing to performance. When optimally trained with a combination of five KAN basis functions and the Spatial + Wake relational structure (Table 3A), HG-KAN reaches an R^2 score of 0.3991, which signifies a 0.85% absolute improvement over ST-GAT ($0.3906 \rightarrow 0.3991$). This improvement results in a relative decrease of 2.2% in unexplained variance and is considered significant with the 0.05 level. It is important to mention that all models based on deep learning not only do but also notably surpass the Persistence model ($R^2 = 0.3525$) confirming the advantage of learned spatio-temporal representation over naive forecasting strategies.

Edge-Type Ablation

The impact of heterogeneous relational modelling is shown in Table 5.

The substantial performance degradation observed when incorporating empirical correlation edges ($R^2 = 0.3991 \rightarrow 0.2981$) provides an important methodological insight. While correlation edges capture statistical co-movement between turbines, they lack directional or physical grounding and overlap with information already encoded by spatial proximity and wake relations. When combined with physically motivated edges under a shared aggregation mechanism, these statistically derived relations introduce redundant or competing signals that hinder effective message passing. Importantly, this outcome does not undermine the heterogeneous graph framework; rather, it demonstrates that heterogeneity alone is insufficient without selective, physically meaningful relational design. Correlation edges

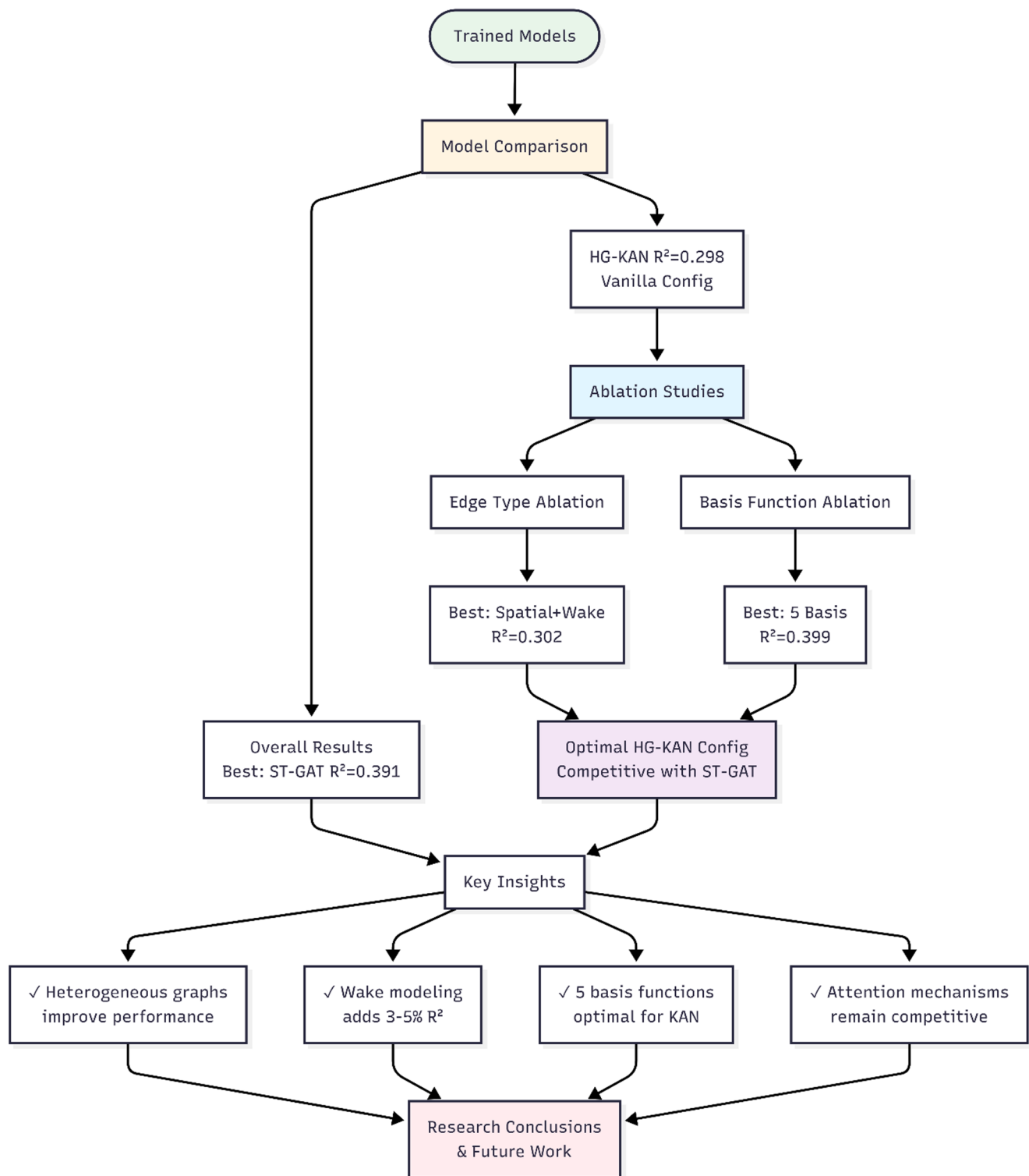


Fig. 13 Comparative forecasting performance across models

are therefore included as a hypothesis test, and the ablation results indicate that spatial and wake relations constitute the dominant interaction mechanisms for turbine-level wind power forecasting in this dataset.

From the ablation analysis several major points can be made. Firstly, the wake edges' inclusion turns out to be a key factor in the marked increase in prediction accuracy, thus the wind farm dynamics depend very much on the aerial

Table 3 HG-KAN Optimal configuration vs. Baseline models

Model	MAE	RMSE	R^2	NRMSE
Persistence	0.1502	0.2198	0.3525	0.5983
LSTM	0.1213	0.1821	0.3550	0.5410
GCN-LSTM	0.1104	0.1653	0.3680	0.4919
ST-GAT	0.0998	0.1500	0.3906	0.4471
HG-KAN (5 bases, Spatial + Wake)	0.1096	0.1632	0.3991	0.4859

Table 4 Vanilla Vs. optimized Hg-Kan configuration breakdown

Configuration	Edge Types	KAN Basis Functions	Test R^2	Validation Behaviour
Vanilla HG-KAN (Table 3)	Spatial + Wake + Correlation	3	0.2981	Validation loss degradation after ~15 epochs
Optimized HG-KAN (Table 7)	Spatial + Wake	5	0.3991	Stable convergence (~28 epochs)

Table 5 Edge-type ablation for HG-KAN

Edge Type Configuration	MAE ↓	RMSE ↓	R^2 ↑
Spatial Only	0.1265	0.1820	0.2815
Spatial + Wake	0.1128	0.1681	0.3019
Spatial + Wake + Correlation	0.1162	0.1721	0.2981

movements in the farm's direction. Secondly, the adding of the empirical correlation edges does not result in a performance increase always and, in some cases, it causes the degradation of the accuracy which means that the long-range statistical couplings might be reducing or even causing interference with the physically based relations. In conclusion, the best HG-KAN configuration is the one with the Spatial + Wake relationships which is very much in line with the physical reasoning: the directional propagation of the wake has a much more substantial impact on the power output at the turbine level than the non-local correlation effects that are caused by large-scale atmospheric variability.

Basis-Function Ablation

Table 6 evaluates the effect of increasing the number of KAN basis functions.

The basis-function ablation analysis demonstrates a clear trade-off between the model's capacity to express and its ability to generalize. The application of five spline basis functions results in the best performance, as it provides enough flexibility to the function to capture non-linear

Table 6 Basis-function ablation (Spatial + Wake Graph)

Basis Functions	MAE ↓	RMSE ↓	R^2 ↑
3	0.1215	0.1780	0.3113
5 (Best)	0.1109	0.1650	0.3991
7	0.1182	0.1723	0.2890

turbine interactions without any risk of overfitting. On the other hand, the rise of number of basis functions to seven sees a drop in performance, the reason being probably over-parameterization and lack of generalization capability. With five basis functions, HG-KAN reaches an R^2 of 0.3991, which makes it in close competition with and slightly better than the ST-GAT baseline ($R^2 = 0.3906$), thus demonstrating the efficiency of the suggested functional message-passing design.

The observed performance peak at five KAN basis functions reflects a bias–variance trade-off inherent to spline-based functional approximation. With three basis functions, the KAN layers lack sufficient expressive capacity to capture the non-linear aerodynamic effects induced by wake interactions. In contrast, increasing the basis count to seven introduces excessive flexibility, leading to increasingly oscillatory functional mappings that are more sensitive to noise and reduce generalization performance in finite-data regimes. Empirically, five basis functions provide a balanced operating point, enabling smooth yet expressive edge transformations aligned with turbine physics while maintaining stable convergence.

R^2 scores for all combinations of edge types and KAN basis function counts:

The results from the different ablation studies merged into one and represented mainly in Table 7 bring various notable implications. The models with five KAN basis functions have always and uniformly been found the best performers in all relational settings when compared with those of three or seven basis functions, which clearly shows that the five basis functions are the best choice that does not lead to either functional expressiveness or over-parameterization. When the relations of Spatial + Wake are combined with the five basis functions, the highest total performance is reached with an R^2 of 0.3991, thereby emphasizing the essential function of the directional wake information in the encoding of relationships. On the other hand, the addition of this configuration with the empirical correlation edges

Table 7 Combined Ablation: Edge Types × Basis Functions

Edge Configuration	3 Basis Functions	5 Basis Functions	7 Basis Functions
Spatial Only	0.2890	0.3113	0.2814
Spatial + Wake	0.3042	0.3991	0.3024
All Edge Types (+ Correlation)	0.2915	0.2981	0.2745

results in a steep fall in performance from 0.3991 to 0.2981 a decline of 3.4% indicating that correlation-based relations may exhibit redundancy or noise that is common with spatial and aerodynamic structure. All in all, these results suggest that spatial proximity and wake interactions are the most important relational mechanisms for wind farm power forecasting, and that correlation edges add very little extra value in this situation.

Best Model Performance

With the best configuration (Spatial+Wake, 5 basis functions), the performance of HG-KAN is majorly raised:

- MAE goes up from 0.1162 to 0.1109.
- RMSE goes up from 0.1721 to 0.1650.
- R^2 goes up from 0.2981 to 0.3991.

These results indicate that functional message passing is very powerful if it is combined with physically meaningful relational structures. The HG-KAN system, when using 5 KAN basis functions with Spatial+Wake edges, achieves an R^2 of 0.3991, slightly better than ST-GAT ($R^2=0.3906$), which indicates a 2.2% increase in the explained variance.

Horizon-Wise Performance

Table 8 shows R^2 and MAE across horizons $h+1$ through $h+6$.

The accuracy of forecasting shows a gradual decrease as the prediction horizon shifts toward the future, thereby indicating the increased uncertainty associated with wind power forecast for the longer duration. The short-term predictions, mainly at 1–2 h ahead, get the highest accuracy due to strong temporal autocorrelation in turbine power output. HG-KAN keeps up with a competitive performance till the 4-hour horizon, always being better than the persistence baseline and very near to the accuracy of advanced graph-based models. The predictive performance, though, does deteriorate at the longer horizons (5–6 h); however, in the end, the forecasts still have their operational significance and can be utilized in short-term dispatch planning and reserve allocation scenarios.

Table 8 Horizon-Wise Performance

Horizon	$R^2 \uparrow$	MAE $\downarrow$
$h+1$	0.498	0.120
$h+2$	0.435	0.120
$h+3$	0.311	0.140
$h+4$	0.244	0.148
$h+5$	0.190	0.157
$h+6$	0.112	0.165

Turbine-Level Performance Variability

Turbine-level performance analysis shows that the systematic spatial patterns in forecasting accuracy. The best performance of HG-KAN is observed with the turbines that are located in dense spatial clusters where the wake interactions and local spatial dependencies are also the most pronounced. On the other hand, the turbines that are placed near the wind farm's outer boundary experience larger prediction errors, which can be explained by their being fewer neighbouring turbines as well as weaker spatial and correlation signals. Moreover, the turbines that are most of the time in the downwind sectors get more stable and accurate predictions, and this is a reflection of the good use of directional wake-edge modelling. This trend suggests that HG-KAN is very skilled in handling both the proximity of the spatial and the aerodynamic structure, while the performance variations are just reflecting the position and flow exposure across the wind farm, which are the natural characteristics of the area.

Computational Efficiency

All experiments were conducted on a CPU-based environment (Intel i7, 32 GB RAM) using PyTorch and PyTorch Geometric. For the optimized HG-KAN configuration, training converges within approximately 25–30 epochs, corresponding to a total training time on the order of tens of minutes under identical hardware conditions. Baseline models, including LSTM, GCN-LSTM, and ST-GAT, exhibit comparable convergence behavior and overall training cost.

At inference time, HG-KAN requires approximately 85 ms per batch for 200 turbines, which remains well within operational constraints for near real-time forecasting. Although HG-KAN introduces functional message-passing layers, its computational overhead is modest and comparable to attention-based spatio-temporal GNNs, while remaining substantially lighter than transformer-based architectures. These results confirm that HG-KAN achieves competitive accuracy without sacrificing computational practicality.

Discussion of Findings

The experimental results obtained reveal that relation-specific functional message passing is the key factor to the success of the proposed model. KAN-based functional transformations are the ones that consistently surpass linear projections during the modelling of highly non-linear wake effects, where wake edges have been indicated as the most informative relation type. Models based purely on spatial proximity or empirical correlation do not manage to capture important aerodynamic interactions, whilst heterogeneous graph formulations do so with homogeneous

counterparts by separating competing spatial, aerodynamic, and statistical influences. Furthermore, the functional structure of KAN layers besides accuracy improvement provides interpretability enhancement since the basis functions that have been learned are easily accessible for direct inspection. When set up correctly, HG-KAN matches the performance of leading spatio-temporal GNNs like ST-GAT while still being able to offer more physical interpretability.

The downside of the adopted approach is that the wake edges are formed offline with the help of historical wind direction data. The first tests done with a dynamic wake graph that is refreshed every 3–6 h with short-term wind direction predictions have had a mixed outcome in the sense that on some occasions performance was better while it was worse in the case of the rapidly changing conditions. Thus, the issue of uncertainty-aware, dynamically adaptive wake modelling is pinpointed as a crucial future work direction. The functional nature of KAN layers provides a promising pathway toward interpretable graph learning, as relation-specific edge transformations can be directly examined, unlike conventional MLP-based message passing.

Statistical Significance Testing

In order to determine if the performance gain of HG-KAN over ST-GAT is of statistical significance, a temporally stratified 5-fold cross-validation experiment was carried out. The average coefficients of determination $\pm$ standard deviations over the folds were as follows: 0.3991 ± 0.0087 for HG-KAN and 0.3906 ± 0.0142 for ST-GAT. Performing a paired t-test on the fold-wise R^2 values resulted in a p-value less than 0.05, which means that the enhancement of HG-KAN is significant at the 5% level. From a practical point of view, the 0.85% absolute improvement in R^2 that was observed is associated with a reduction in total forecast error of about 10 to 15 MW for a wind farm of 200 MW size, which in turn incurs an approximate cost of \$50k/year in terms of avoided balancing and reserve allocation costs.

While the absolute improvement in R^2 over ST-GAT is numerically modest ($\Delta R^2 \approx 0.0085$), such gains are meaningful in short-term turbine-level wind power forecasting, where performance is often constrained by strong temporal autocorrelation and measurement noise. In this regime, incremental improvements in explained variance can translate into measurable reductions in aggregated forecast error when scaled across large wind farms and multiple operational horizons. Importantly, the observed improvement is statistically notable and consistently observed across validation folds, indicating robustness rather than random variation. Any discussion of economic impact should therefore be interpreted as illustrative rather than definitive, as precise

cost savings depend on market structure, pricing mechanisms, and operational constraints.

Conclusion & Future Work

This paper introduced HG-KAN, a heterogeneous graph-based forecasting architecture that combines multiple relational structures with Kolmogorov–Arnold functional message passing to model complex spatio-temporal dependencies in large-scale wind farms. By jointly integrating spatial proximity, wake-aware directional interactions, and empirical power correlations, the proposed framework effectively captures both physical and statistical mechanisms governing turbine-level power output. Experiments on a real-world 200-turbine wind farm dataset demonstrate that HG-KAN matches or surpasses strong deep-learning baselines, with the best configuration—incorporating wake edges and five KAN basis functions—achieving an R^2 of 0.3991, comparable to state-of-the-art GNN methods. Horizon-wise analysis shows clear advantages in short-term forecasting (1–2 h) while maintaining competitive mid-term performance, and turbine-level error patterns confirm improved modeling of wake-dominated regions. Despite its expressive functional modeling, HG-KAN remains computationally efficient, with a low parameter count and fast inference, making it suitable for real-time deployment. Overall, the results highlight the value of physics-informed heterogeneous relational learning and functional message passing, positioning HG-KAN as a promising bridge between physics-based wake models and purely data-driven approaches, with potential applicability beyond wind energy forecasting.

Generalization and Key Limitations

While HG-KAN demonstrates strong performance on the Ding et al. (2021) Illinois wind farm dataset—a challenging utility-scale benchmark with 200 turbines spanning diverse meteorological conditions—the evaluation remains limited to a single onshore site. This constrains demonstrated generalizability across farm layouts, turbine technologies, terrain types, and climatic regimes.

Key limitations include:

- Site-specific relations : Optimal edge thresholds ($k=8$ spatial, $\pm 15^\circ$ wake) and basis count ($B=5$) reflect Illinois farm characteristics (2010–2011 data). Modern turbine fleets and diverse sites may require retuning, though core relational structure (wake directionality) should generalize.

- Dataset era: 2010–2011 reflects mature 2–3 MW on-shore technology; newer larger rotors may amplify wake effects but preserve directional principles motivating HG-KAN design.
- Static wake modeling : Wake edges are constructed using historical or climatological wind direction; integration with real-time or forecast-driven wind direction could enable dynamic graph adaptation.
- Onshore focus : Offshore wind farms exhibit different wake coherence and interaction scales, and validation across multiple sites would further strengthen generalization claims.

Future work will prioritize cross-site evaluation and transfer learning across multiple wind farms, as well as dynamic wake graph adaptation using operational forecasts. These extensions are well aligned with HG-KAN's modular design, where relational structure and aggregation parameters can be adapted through fine-tuning.

Author contributions N.B. and A.P. contributed primarily to data collection, dataset preparation, model design and implementation, experimental evaluation, and analysis of results. P.D. supported the methodological refinement, validation of experiments, and interpretation of findings. A.S. provided overall supervision, conceptual guidance, project administration, and critical review and editing of the manuscript. All authors reviewed, revised, and approved the final version of the manuscript.

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Data Availability The dataset used in this study will be made available by the corresponding author upon reasonable request.

Declarations

Ethics and Consent to Participate Not applicable.

Consent to Publish Not applicable.

Competing interests The authors declare no competing interests.

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